

Main Checklist for Data_Project

Progress tracking for the DDR Data Logger project on Zybo Z7-20

Main Checklist

Stage	Stage Name	Status	Stage Output
0	GitHub + Project Structure	✓ Done	Repository, folders, README, docs, first commit
1	Data Protocol	✓ Done	docs/data_protocol.md with Header, Length, Data, Checksum
2	Data Generator + Testbench	✓ Done	rtl/data_generator.v, tb/tb_data_generator.v, simulation, waveform, report
3	Packet Builder FSM + Testbench	✓ Done	packet_builder_fsm.v + testbench + simulation
4	Sync FIFO + Testbench	✓ Done	sync_fifo.v + testbench + simulation
5	AXI-Stream Master + Testbench	✓ Done	axis_stream_master.v + testbench + simulation
6	Full RTL Integration Simulation	✓ Done	tb/tb_data_project_integration.v + full RTL integration simulation + waveform + console PASS
7	Custom IP Packaging + Vivado Block Design	☐ Pending	Separate custom IP blocks and visual Block Design prepared for AXI DMA / DDR
8	Vitis C Verification	☐ Pending	C software that starts DMA and verifies DDR data
9	Async FIFO + CDC + Testbench	☐ Pending	async_fifo.v + testbench with two clocks
10	Full Integration with CDC + DMA + DDR	☐ Pending	Full system with two clock domains and DDR
11	ILA + Vivado Reports	☐ Pending	ILA, timing report, utilization report
12	Python CSV / Graph	☐ Pending	CSV, graph, and documentation
13	UART or SPI Extension	☐ Pending	Real data source replacing Data Generator
14	Portfolio Polish	☐ Pending	GitHub, README, images, explanations, and reports polish

Current status: Stages 0-6 are complete. The next stage is Stage 7 - Custom IP Packaging + Vivado Block Design.

Detailed Checklist for Work Tracking

Stage 0 - GitHub + Project Structure

✓ Done

- ✓ Created Data_Project folder
- ✓ Created GitHub repository
- ✓ Created README.md
- ✓ Created organized folder structure
- ✓ Created work plan document
- ✓ Added block diagram
- ✓ Completed first commit
- ✓ Completed push to GitHub

Stage 1 - Data Protocol

✓ Done

- ✓ Created docs/data_protocol.md
- ✓ Defined DATA_WIDTH = 32
- ✓ Defined Header
- ✓ Defined Length
- ✓ Defined Data sequence
- ✓ Defined Checksum
- ✓ Defined Expected Packet
- ✓ Defined PASS / FAIL Criteria
- ✓ Completed commit
- ✓ Completed push

Stage 2 - Data Generator + Testbench

✓ Done

- ✓ Defined block goal
- ✓ Defined inputs and outputs
- ✓ Wrote rtl/data_generator.v
- ✓ Wrote tb/tb_data_generator.v
- ✓ Ran Behavioral Simulation
- ✓ Verified expected sequence: 1, 7, 8, 3, 6, 3
- ✓ Saved waveform screenshot: images/stage2_data_generator_waveform.png
- ✓ Created simulation report: reports/stage2_data_generator_simulation.md
- ✓ Completed commit
- ✓ Completed push

Stage 3 - Packet Builder FSM + Testbench

✓ Done

- ✓ Define block goal
- ✓ Define FSM states
- ✓ Write rtl/packet_builder_fsm.v
- ✓ Write tb/tb_packet_builder_fsm.v
- ✓ Run simulation
- ✓ Verify Header / Length / Data / Checksum
- ✓ Complete commit
- ✓ Complete push

Stage 4 - Sync FIFO + Testbench

✓ Done

- ✓ Define block goal
- ✓ Write rtl/sync_fifo.v
- ✓ Write tb/tb_sync_fifo.v
- ✓ Verify write/read
- ✓ Verify full/empty
- ✓ Verify overflow/underflow
- ✓ Complete commit
- ✓ Complete push

Detailed Checklist for Work Tracking

Stage 5 - AXI-Stream Master + Testbench

✓ Done

- ☒ Define block goal
- ☒ Write rtl/axis_stream_master.v
- ☒ Write tb/tb_axis_stream_master.v
- ☒ Verify tvalid / tready / tdata / tlast
- ☒ Verify backpressure
- ☒ Complete commit
- ☒ Complete push

Stage 6 - Full RTL Integration Simulation

✓ Done

- ☒ Reviewed RTL block interfaces
- ☒ Added data_ready to packet_builder_FSM
- ☒ Wrote tb/tb_data_project_integration.v
- ☒ Connected Data Generator → Packet Builder FSM → Sync FIFO → AXI-Stream Master
- ☒ Ran full RTL integration simulation
- ☒ Verified expected AXI packet words
- ☒ Verified axis_tlast on checksum word
- ☒ Saved waveform screenshot and console PASS
- ☒ Complete commit
- ☒ Complete push

Stage 7 - Custom IP Packaging + Vivado Block Design

□ Pending

- ☐ Package data_generator as IP
- ☐ Package packet_builder_FSM as IP
- ☐ Package sync_fifo as IP
- ☐ Package axi_stream_master as IP
- ☐ Add custom IP repository
- ☐ Create Block Design
- ☐ Connect custom IP blocks visually
- ☐ Prepare AXI DMA / DDR integration
- ☐ Complete commit
- ☐ Complete push

Stage 8 - Vitis C Verification

□ Pending

- ☐ Create Vitis project
- ☐ Write DMA initialization code
- ☐ Define DDR buffer address
- ☐ Start DMA transfer
- ☐ Read data from DDR
- ☐ Verify Header / Length / Data / Checksum
- ☐ Complete commit
- ☐ Complete push

Stage 9 - Async FIFO + CDC + Testbench

□ Pending

- ☐ Write rtl/async_fifo.v
- ☐ Write tb/tb_async_fifo.v
- ☐ Verify two clocks
- ☐ Verify correct data crossing
- ☐ Complete commit
- ☐ Complete push

Stage 10 - Full Integration with CDC + DMA + DDR

□ Pending

- ☐ Integrate Async FIFO into the system
- ☐ Update full integration
- ☐ Verify two clock domains
- ☐ Verify DDR write
- ☐ Complete commit
- ☐ Complete push

Detailed Checklist for Work Tracking

Stage 11 - ILA + Vivado Reports

☐ Pending

- ☐ Add ILA
- ☐ Verify key signals
- ☐ Save timing report
- ☐ Save utilization report
- ☐ Save screenshots
- ☐ Complete commit
- ☐ Complete push

Stage 12 - Python CSV / Graph

☐ Pending

- ☐ Create CSV file
- ☐ Write Python script
- ☐ Create graph
- ☐ Save screenshot / output
- ☐ Complete commit
- ☐ Complete push

Stage 13 - UART or SPI Extension

☐ Pending

- ☐ Choose UART or SPI
- ☐ Write communication block
- ☐ Run verification
- ☐ Receive real data through the system
- ☐ Complete commit
- ☐ Complete push

Stage 14 - Portfolio Polish

☐ Pending

- ☐ Update README
- ☐ Add images
- ☐ Add waveforms
- ☐ Add reports
- ☐ Add explanations for each block
- ☐ Add Future Work / Lessons Learned
- ☐ Prepare project for presentation